



SERIES OF EXERCISES N° 1 : REAL NUMBERS

Exercise 1

1. Let $n \in \mathbb{N}^*$ such that n is not the square of any integer. Show that $\sqrt{n} \notin \mathbb{Q}$.
2. Prove that $\sqrt{2} + \sqrt{3} \notin \mathbb{Q}$.

1. Inequalities

Exercise 2

I) Let $x, y, z \in \mathbb{R}$. Prove the following inequalities :

1. $xy + yz + xz \leq x^2 + y^2 + z^2$,
2. $|\sqrt{|x|} - \sqrt{|y|}| \leq \sqrt{|x - y|}$,
3. $\frac{x + y}{2} \geq \sqrt{xy}$ for all no-negative real numbers x and y .

II) Prove the Cauchy Schwarz's inequality

For all $n \in \mathbb{N}^*$, $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$, be no-negative real numbers, we have

$$\left(\sum_{i=1}^n x_i y_i \right)^2 \leq \left(\sum_{i=1}^n x_i^2 \right) \times \left(\sum_{i=1}^n y_i^2 \right).$$

2. Reasoning by Induction

Exercise 3

Prove by induction the following assertions.

1. $\forall n \in \mathbb{N}^*, \sum_{k=1}^n k^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$
2. $\forall n \in \mathbb{N} \quad |\sin(nx)| \leq n |\sin x|, \quad \forall x \in \mathbb{R}$
3. Show that, for all $n \in \mathbb{N}^*$

$$\sum_{k=1}^n \frac{1}{\sqrt{k}} < \sqrt{n} + \sqrt{n+1} - 1.$$

3. The integer part function

Exercise 4

We denote by $E(x)$ or $\lfloor x \rfloor$; the integer part of x . Prove the following results

1. $* E(x) + E(-x) = \begin{cases} 0 & \text{if } x \in \mathbb{Z}. \\ -1 & \text{if } x \in \mathbb{R}/\mathbb{Z}. \end{cases}$
2. $* \forall x, y \in \mathbb{R}, x \leq y \Rightarrow E(x) \leq E(y)$
3. $\forall x, y \in \mathbb{R}, E(x) + E(y) \leq E(x + y) \leq E(x) + E(y) + 1$
4. Show that, for all $n \in \mathbb{Z}$

$$\left\lfloor \frac{n-1}{2} \right\rfloor + \left\lfloor \frac{n+2}{4} \right\rfloor + \left\lfloor \frac{n+4}{4} \right\rfloor = n.$$

4. Upper bound sets , Lower bound sets

Exercise 5

Let A and B be two nonempty bounded sets of real numbers. Show that

1. If $A \subset B$ then, $\inf B \leq \inf A$ and $\sup A \leq \sup B$
2. $\sup(A \cup B) = \max(\sup A, \sup B)$, $\inf(A \cup B) = \min(\inf A, \inf B)$,

Exercise 6 (Using characterisation of sup , inf)

1) Let A and B be two nonempty bounded sets of real numbers. Let

$$C = A + B = \{a + b, (a, b) \in A \times B\}$$

Show that C is a bounded set and that $\sup C = \sup A + \sup B$ and $\inf C = \inf A + \inf B$.

2) Assume that A is nonempty and bounded above set of no-negative real numbers. Prove that $\sup(\sqrt{A}) = \sqrt{\sup A}$.

Exercise 7 Find if they exist, the infimum and supremum of each of the following sets.

$$\begin{aligned} A &= \{x \in \mathbb{N}, x^2 - x - 6 > 0\}, \quad B = \{n \cos(n\pi); n \in \mathbb{N}\}, \quad *C = \{x \in \mathbb{Q}, x^3 \leq 8\} \\ *D &= \left\{ \frac{(-1)^n}{n} e^{-n}, n \in \mathbb{N}^* \right\}, \quad E = \left\{ \frac{n}{mn+1}; (m, n) \in \mathbb{N}^{*2} \right\}, \quad F = \{\sqrt{n} - E(\sqrt{n}), n \in \mathbb{N}\} \\ G &= \left\{ \frac{m}{n} + \frac{4n}{m}, m, n \in \mathbb{N}^* \right\}, \quad H = \left\{ 2^x + 2^{\frac{1}{x}}, x \geq 0 \right\}, \end{aligned}$$

5. Density in $\mathbb{R}$

Exercise 8

Prove that

1. The set of dyadic rationals denoted by $\mathcal{D}$ such that :

$$\mathcal{D} = \left\{ \frac{a}{2^n}, a \in \mathbb{Z}, n \in \mathbb{N} \right\}$$

is dense in $\mathbb{R}$.

2. ** The set of decimal numbers is dense in $\mathbb{R}$.